

USRP X310 Data Recording Report

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Last update: October 8, 2026

Abstract

This report describes the data recording process using the USRP X310 device. This receiver is placed in a fixed position in São José dos Campos, SP, Brazil, in the GPS/GNSS Laboratory at Aeronautics Institute of Technology (ITA)¹. The present report is updated as new recordings are made and new information is available.

1 Overview of the recordings

The recording is computed automatically by a Linux service which is driven by the following guidelines:

- Everyday, the program will start within the time range [4PM, 6PM]
- It will sense whether the USRP is detectable via `uhd.find_devices --args='addr=192.168.10.2'`.
- If not found, it will exit silently and will sense again in intervals of 15 min.
- Once found, a SigMF file will be created from the recording, with the interleaved I/Q samples + metadata, like number of channels (dual channel recording), frequency (L1 and L5), sampling rate (2MHz), sample type (signed 16-bit int), etc.
- Each record lasts 5 minutes (300 s)
- It will record a second time within the same range and 30 min after the first record, and then it will not record anymore for that day.
- After 15 days of records from the last report. Rubem will send an automatic message to header of the Laboratory, Prof. Dr. Felix Antreich, via WhatsApp with the summarized information from all recorded metadata;

¹<https://github.com/ita-gnss-lab>

October						
M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Table 1: Recording calendar. **Green**: 2 recordings made; **blue**: 1 recording made. No color: no recording made.